
Chandler McArthur's Goodreads Profile Analysis

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December 2022



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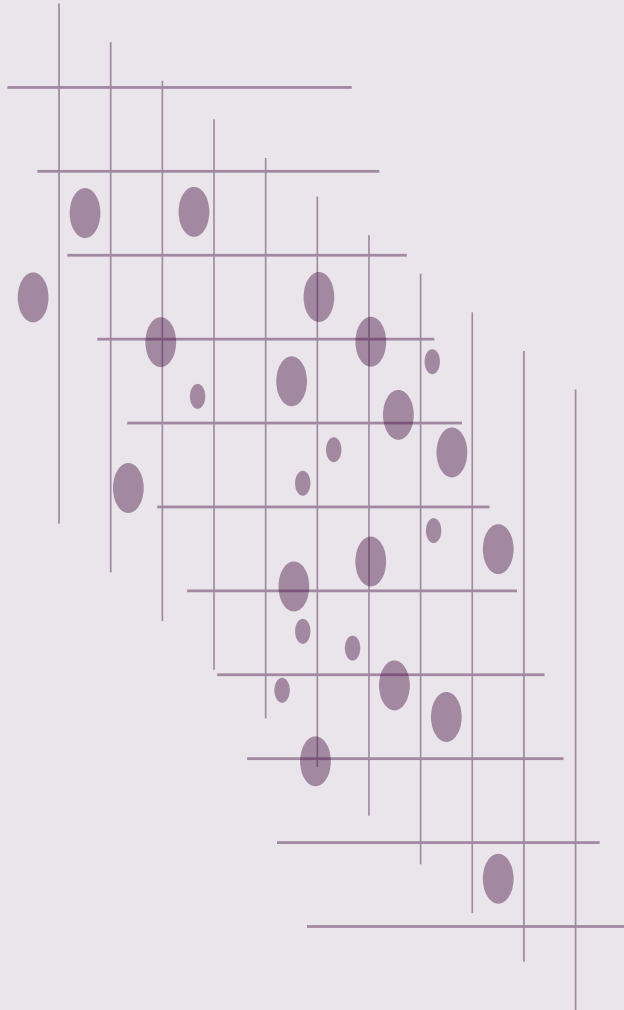
01 Overview

Chandler's Goodreads profile offers a wealth of data that can be analyzed to inform future literature choices. While [the Goodreads website](#) itself provides some basic evaluation tools, this Christmas gift is intended to delve deeper and enhance Chandler's reading experience.

Considering Chandler's strong appreciation for reading, this presentation aims to enrich his enjoyment of the hobby.



02 Methodology



Up until December 6, 2022, data from Goodreads was manually entered into an Excel spreadsheet. Visualizations were created using Tableau, and predictive analytics were conducted using Python.

In some cases, data was not available on Goodreads and had to be sourced from Amazon metrics and online research to fill in missing fields.

Data Dictionary

Name	Description	Type	Example
ID	Unique ID of book	Integer	34
Title	Title of book	Text	"Dracula"
Author	Author of book	Text	Bram Stoker
Avg_Rating	Average rating of book on Goodreads	Float	4.01
Chandler_Rating	Chandler's rating (out of 5) of the book	Integer	3
Reviewed	Whether Chandler reviewed the book	Boolean	Y
Graphic_Novel	Whether book was a graphic novel	Boolean	N
Fiction_NonFiction	Whether book was nonfiction or fiction	Text	Fiction
Pages	Number of pages in the book	Integer	488
Published_Year	Year book was first published	Integer	1897
Year_Added	Year Chandler added book to Goodreads	Integer	2020
Genre	Genre of the book, taken from Chandler's own personal list	Text	Lit

03 Analysis/Visualizations

Descriptive Statistics

	Avg_Rating	Chandler_Rating	Pages	Published_Year	Year_Added
count	121.0				
mean	3.939917	4.247934	256.115702	1932.380165	2020.239669
std	0.359199	0.924491	140.451604	273.202070	1.329316
min	2.0	1.0	20.0	439.0	2018.0
25%	3.78	4.0	160.0	1974.0	2019.0
50%	3.98	4.0	226.0	2005.0	2020.0
75%	4.17	5.0	336.0	2015.0	2021.0
max	4.75	5.0	719.0	2021.0	2022.0

Regression Outputs

Logistic Regression Model - Review Likelihood

```
Chandler_Rating: -0.597
Pages: 0.005
Avg_Rating: -0.318
Published_Year: -0.002
NonFiction: 0.075
Graphic_Novel: -0.881
-----
Intercept: 7.836
```

R-squared: 0.68

- High ratings paradoxically reduce likelihood of review
- Longer books increase review likelihood
- Model explains 68% of review behavior

Regression Outputs

Multinomial Logistic Regression Model - Ratings

Intercept	Avg_Rating	Graphic_Novel	Nonfiction	Pages	Published_Year	Year_Added	Predicted Rating
-0.00104	0.028	-0.051	0.073	-0.0414	0.190	-0.188	1
-0.00012	-0.721	-0.203	0.204	0.0037	-0.053	0.054	2
0.00059	-1.501	0.101	0.896	0.0110	-0.032	0.034	3
0.00213	0.195	0.153	-0.295	0.0133	-0.052	0.051	4
-0.00154	1.999	0.000	-0.877	0.0134	-0.053	0.049	5

R-squared: 0.48

- Avg rating is the strongest predictor: higher-rated books → higher predicted rating
- Model captures 48% of rating variation (weaker predictability than review model)

[Click here for Tableau V
isualizations](#)

Observations

Deviation intensifies for negative reviews. Chandler rarely completely aligns with the community– he both likes and dislikes more than the average sentiment.	Books published in 20th and 21st centuries are most read. Increase in # of pages seem to correlate slightly with higher rating but rating itself is not affected by time period.
Reviews are a bit of a black box. Likelihood of reviews increases with pages, and potentially with graphic novels and higher ratings.	Classics, Franchise, and Lit genres have the largest deviation in rating. Chandler tends to favor all genres more in ratings given, except for philosophy/psych (which could be outliers; more data is needed).
Fiction is more well-read and well-loved. More nonfiction data might be needed to gauge an accurate reading.	Classics, Franchise, and Lit genres also are the most well-read. Franchise genre needs formal definition to serve as official new genre in list.

04 Recommendations

Insights



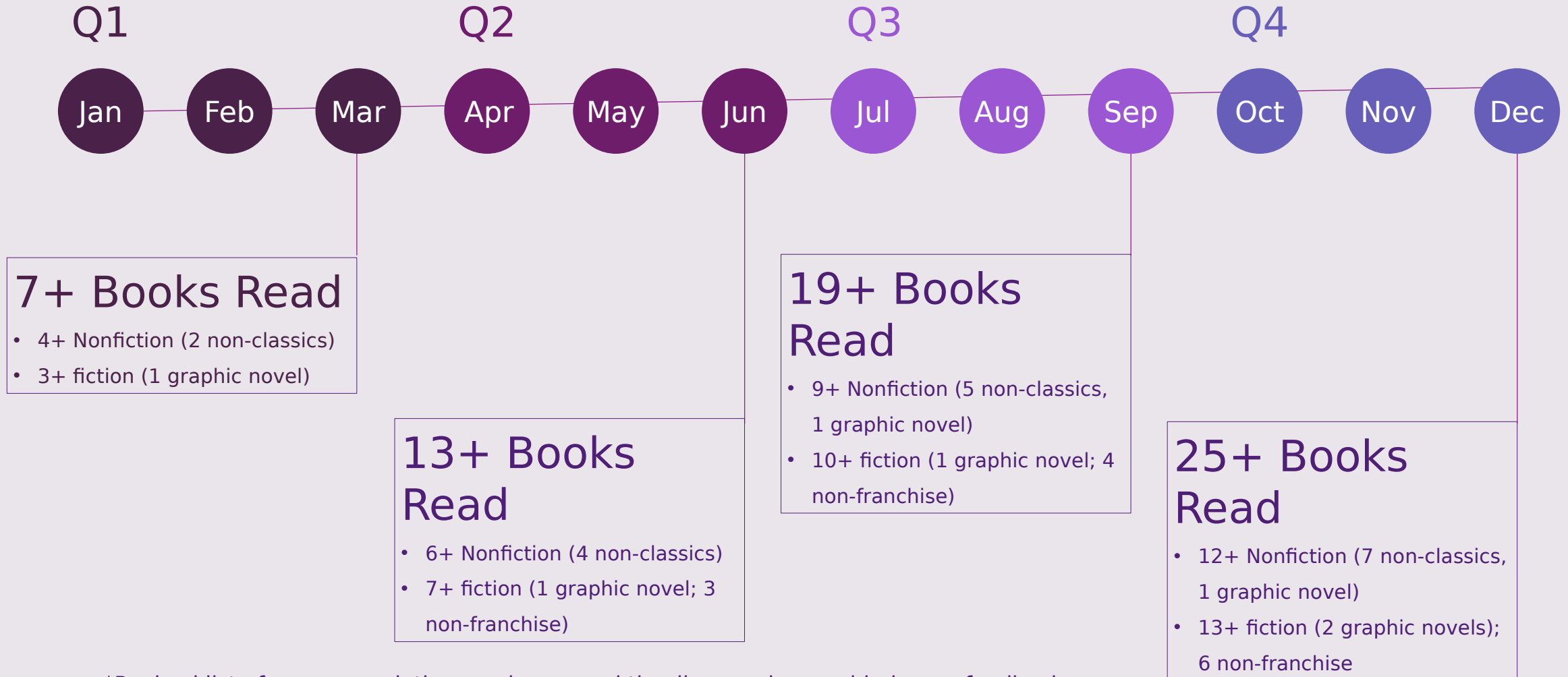
Suggestions

- Genre and fiction/nonfiction designation seem to be more important factors in selecting a book than the Goodreads average rating
- Attempt more books from ancient/past eras
- Read more varied nonfiction
- Graphic novels are generally a safe bet
- Rework genre list to account for overlaps and nuances

Assumptions

- Same metrics for goal: 25 books by end of 2023
- Aiming for more diversity in books read
- Willingness to share data to optimize model calibration

Proposed Timeline



05 Closing

It is our hope that this presentation has served as a useful resource for a deeper understanding of Chandler's reading preferences and habits.

Should you require copies of the materials presented, please do not hesitate to request them.

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Q&A

